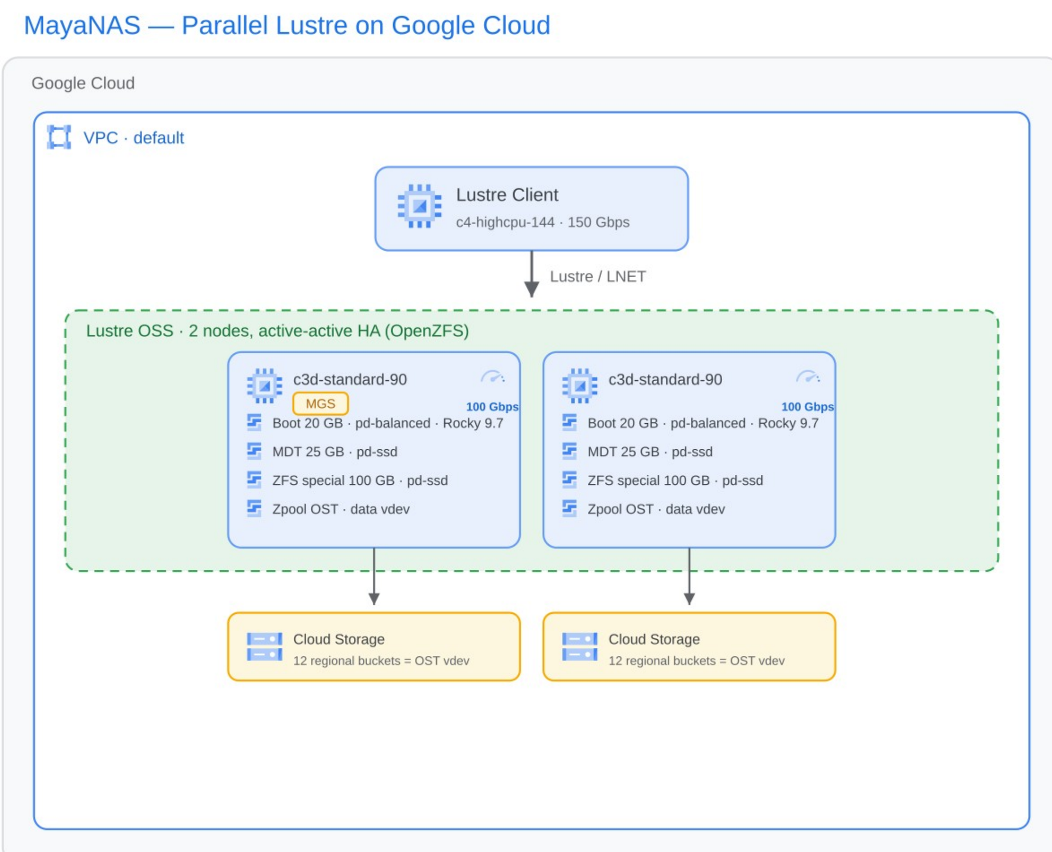
MayaNAS Lustre — MLPerf Storage v3.0 (Checkpointing + KVCache)

Submitter: ZettaLane Systems · **Division:** Closed · **System:** mayanas-lustre-gcp_c4hc144_c3d90_1c_2s

architecture



MayaNAS Storage Cluster (2 nodes, active-active HA)

- 2 × GCP c3d-standard-90 (AMD EPYC 9B14 Turin, 90 vCPU / 45 cores, 354 GB RAM)
- Rocky Linux 9.7 · 100 Gbps Tier_1 networking
- Each node: Lustre OSS with 1 OST (OpenZFS zpool; data vdev = 12 Standard regional GCS buckets via objbacker.io)
- Per node: Boot 20 GB pd-balanced · MDT 25 GB pd-ssd · ZFS special 100 GB pd-ssd

- MGS on node 1's VIP · regional buckets enable HA failover (surviving node imports partner zpool)

Benchmark Client

- 1 × GCP c4-highcpu-144 (Intel Xeon 6985P-C (Granite Rapids), 144 vCPU, 283 GB RAM)
- Ubuntu 24.04 · 150 Gbps networking
- Lustre client mount (hybrid_io=1)

Results

- Checkpointing (llama3-8b, np=8): save ~14.4 GB/s, load ~10.4 GB/s (Pattern A, 10 writes + 10 reads)
- KVCache (llama3.1-8b, stress): 525 tok/s, 9.14 GB/s read

Availability

MayaNAS and MayaScale can be deployed from the Google Cloud Marketplace portal or as Terraform modules (https://github.com/zettalane-systems/open-lustre-cloud). Contact: info@zettalane.com.